

4	Computer Graphics & 3D Game Engines	VR	
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4. Leaning Out comes

Throughout this course, I have gained a solid understanding of geometry and its significance in the theory part. Studying the dot product has taught me how to determine whether two vectors are parallel, antiparallel, or perpendicular, and to apply this knowledge in real-world scenarios, such as in manual-handling operations. Exploring the cross product revealed how to generate a vector orthogonal to two given vectors, a concept that underpins analyses of rotational forces in mechanical engineering. Furthermore, the scalar triple product offers a straightforward criterion for testing whether three vectors are coplanar, which is invaluable for computing volumes and verifying spatial configurations. In conclusion, although the material initially appeared abstract, I now appreciate the depth and practical power of geometric methods.

5. Exercises

Last section presents the answers of the most important exercises which have been uploaded on the GitHub as different solutions for a further generation of students.